Test Plan: RAG-Diary

> The system test plan: four test levels, the environment, test cases TC-01 to TC-24, and the entry and exit criteria, so testing is systematic and traceable.

> Thai version: TEST-PLAN.md

1. Document control

Item	Detail
Document	Test Plan
Version	1.1
Prepared	15 October 2026
Author	Business Analyst / System Analyst (project owner)
Status	Reviewed, round 1 (25 October 2026); test evidence added 14 November 2026
Related documents	SRS, NFR-SPEC, TRACEABILITY-MATRIX, USE-CASES

2. Purpose and scope

This document explains how the system is tested, to confirm it meets its requirements and the quality agreed.

Scope: strategy, test levels, environment, the list of test cases, and entry and exit criteria. Out of scope: the detail of each quality requirement (in NFR-SPEC) and the actual run results (in the UAT report).

3. Test strategy

Testing is risk-based, starting with what affects security, privacy and cost: per-user data isolation and authentication.

Automated testing: unit and API contract tests use bun test. There are 22 tests today (13 server, 9 frontend) and they run in CI on every push alongside typecheck and build. Full end-to-end browser testing still needs Playwright, which is not installed in the project, so system testing is manual this round, with smoke tests against production.

Manual testing: accessibility inspection, load testing and cross-browser testing.

4. Test levels

Level	What is tested	Tool	Owner
Unit	Single functions and modules, such as type validation and context formatting	bun test	Dev
Integration	API working with the database and the AI provider	bun test, simulated environment	Dev
System	Full paths from screen to database	Playwright	QA
Acceptance	Real users confirm the requirements are met	Manual demonstration	PO, QA

5. Test environment

Component	Detail
Frontend	Next.js at http://localhost:3000
Backend	Elysia at http://localhost:3001 (the frontend calls its own /api and Next.js proxies there via API_PROXY_TARGET)
Database	PostgreSQL with pgvector
External service	Google Gemini for embeddings, chat and images
Test data	Two accounts, to test data isolation

6. Test cases

All cases start as Not started until they are actually run.

ID	Description	Verifies	Level	Type	Status
TC-01	Register succeeds, and is refused for an invalid email or a password shorter than 6 characters	FR-01	System	Automated	Not started
TC-02	Login succeeds, a wrong password is reported, and an expired access token is renewed with the refresh token	FR-02	System	Automated	Not started
TC-03	Logout discards the token	FR-03	System	Automated	Not started

ID	Description	Verifies	Level	Type	Status
TC-04	A new entry is written with a tag	FR-04	System	Automated	Not started
TC-05	The semantic index is built automatically when an entry is created	FR-05	Integration	Automated	Not started
TC-06	The user's own entries are listed, newest first	FR-06	System	Automated	Not started
TC-07	Editing an entry rebuilds its index	FR-07	System	Automated	Not started
TC-08	Deleting an entry removes it from the list	FR-08	System	Automated	Not started
TC-09	Semantic search returns related results	FR-09	System	Automated	Not started
TC-10	Chat answers are grounded in the user's entries	FR-10	System	Automated	Not started
TC-11	A mood summary and an image prompt are returned	FR-11	System	Automated	Not started
TC-12	An image is generated from the prompt and its URL is returned	FR-12	System	Manual	Not started
TC-13	Exporting returns a complete file	FR-13	System	Automated	Not started
TC-14	Deleting all data leaves nothing behind	FR-14	System	Automated	Not started
TC-15	Two users cannot see each other's data	FR-15	System	Automated	Not started
TC-16	Search time stays within 3.5 seconds at p95	NFR-01	System	Manual (load)	Not started
TC-17	No cross-user access is found and every endpoint requires authentication	NFR-02	System	Automated	Not started
TC-18	Consent is requested before data leaves, and data can be managed	NFR-03	System	Manual	Not started
TC-19	Every flow works by keyboard and status messages are readable	NFR-04	System	Manual	Not started
TC-20	The system stays usable when an external service fails	NFR-05	Integration	Manual	Not started
TC-21	Response time stays stable as data grows	NFR-06	System	Manual (load)	Not started

ID	Description	Verifies	Level	Type	Status
TC-22	Type checking passes and no data access code is duplicated	NFR-07	Unit	Automated	Not started
TC-23	Rendering is correct across browsers and screen sizes	NFR-08	System	Manual	Not started
TC-24	When the image provider quota is exhausted, the page states the reason and offers the on-device mood card	FR-12, NFR-05	Both	Manual	Not started

First run (25 October 2026): parts of TC-01 and TC-02 passed (short password and malformed email were refused). Cases that need the database and the AI provider could not run because the Supabase project for the test environment was unavailable (the host name did not resolve), so their status stays Not started until the environment is restored.

Latest production check (5 November 2026): API-level smoke tests passed for TC-01 to TC-04, TC-09, TC-13 and TC-14 (register, login, current user, create an entry, search, export and delete all). TC-15 and the remaining cases still need a formal run.

Rate limit check (12 November 2026): seven calls to /ai/imagine inside one minute returned five 503 responses (the image quota is exhausted) and 429 with Retry-After: 60 on calls six and seven, which shows the per-user counter is enforced as designed.

Concurrency check (12 November 2026): concurrent searches at the database layer gave p95 2.78 s at 10 concurrent users and p95 3.51 s at 30. All 400 requests succeeded (200 every time).

Concurrency rerun at 20 users (14 November 2026): p50 1.39 s, p95 10.49 s, and three HTTP 500 responses out of 200 requests. That is worse than the 30-user run, which returned no errors at all, so the problem is stability under high concurrency rather than latency growing with data size. The database connection pool limit has to be checked before TC-21 can be closed.

Every route rate limited (14 November 2026): GET /notes returns 429 on request 61 and DELETE /notes/:id on request 31, both with Retry-After: 60. CI also runs three Playwright smoke tests and two axe accessibility tests on every push.

Per-route counter (14 November 2026): the counter was originally keyed on the user alone, so ten searches also consumed the image quota. Adding the route to the key separates the counters, and the code falls back to the original counter while the migration is unapplied, so deploying first cannot switch the limits off.

Agent-as-user run (Microfish, 14 November 2026): two personas across five scenarios, driven by a model that reads the screen. The agent clicked the submit button repeatedly while it was already working, and failed the sign-up step in the error_recovery scenario (stopped at 20 steps, 0 entries). That points at a button that should be disabled while it waits, not at a conclusion about real users, because this is one simulated session and the agent can overstate what it did.

Test data hygiene (14 November 2026): the agent transcripts are committed as evidence and carried the test-account password, 24 occurrences across 10 files. Credentials are now filtered before the file is written, the existing files are scrubbed, and a script deletes the test accounts. The lesson is to filter at write time, not at commit time.

7. Entry and exit criteria

Entry criteria: requirements are stable, the environment is ready, test data is ready, and the build passes type checking.

Exit criteria: 100 percent of test cases run, a pass rate of at least 95 percent, every Must requirement passing, at least 80 percent of Should requirements passing, no open critical or high defects, and TC-16 to TC-24 all passing.

7.1 Latest performance measurement (5 November 2026)

Measured on production (rag-diary-api.vercel.app) with one test account, 30 entries and 20 Thai queries.

Measure	Result
p50	2.39 seconds
p95	1.31 seconds (load test 12 Nov 2026: 5,000 entries, 100 requests)
max	3.32 seconds
Against NFR-01 (p95 within 3.5 seconds)	Pass

Most of the time is the external embedding call, which sits on every query. That time has not yet been separated from the database search time, so it has to be measured separately before deciding on an architecture change or a further revision of the NFR target.

8. Defect management

Severity	Meaning	Example
Critical	The system is unusable or data leaks	One user can reach another user's data
High	A core function fails	An entry cannot be created
Medium	A failure with a workaround	A status message is unclear
Low	Minor impact	The layout is untidy

9. Roles and responsibilities

Role	Responsibility
Dev	Unit and integration testing, fixing defects
QA	System testing and collecting results
PO	Acceptance testing and the decision to accept

10. References

Document	Relationship
TRACEABILITY-MATRIX	The origin of test case IDs TC-01 to TC-24
NFR-SPEC	The pass criteria for the quality checked by TC-16 to TC-24
RISK-REGISTER	The risks that set the testing focus
USE-CASES	The main and alternative paths written into test cases